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Cold Related Injuries and the Emergency Treatment of Hypothermia

Principles and Practice of Osteopathic Medicine I

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Do.
Make.
Heal.
Innovate.
Reinvent the Future.

Session Objectives: At the conclusion of this lecture the student will be able to:

1. Distinguish the pathophysiology and treatment of the following cold injuries: trench foot; chilblains (pernio); panniculitis; cold urticaria; frostbite
2. Assess and manage patients in various stages of accidental hypothermia
3. Identify the poor prognostic signs which justify termination of resuscitative efforts in accidental hypothermia

Trench Foot

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- Prolonged cooling (***without freezing***) exacerbated by moisture
- May result in peripheral nerve damage
- Initial exam: macerated = waterlogged. (*This is NOT the same as sympathetic pruning such as when you've been in the shower too long*); pale; mottled; anesthetic; and/or pulseless.
- After rewarming: extreme pain; hyperemic; edematous
- Possible permanent sequelae: anesthesia; paresthesia (calambre); dysesthesia; hyperhidrosis; cold sensitivity; gangrene
- Rx: elevation; dry, loose dressing; late debridement

Image from: http://www.theprepperjournal.com/2015/06/03/bug-out-nightmare-stop-trench-foot-before-it-stops-you/

Chilblains (AKA “Pernio”)

- Exposure to intermittent long-term, damp, ***non-freezing*** conditions
- Common in UK and with Raynaud’s phenomenon
- Early: paresthesia; dysesthesia; anesthesia.
- 12-24 hours: pruritus; erythema; edema; cyanosis; nodules; vesicles; ulcerations
- Rewarming may result in the formation of tender blue nodules, which may persist for several days.
- Rx: supportive
- Nifedipine has shown mixed results but is recommended*
- Other meds that have been tried:, NSAIDs, pentoxifylline, topical steroids

https://www.healthline.com/health/chilblains

*UpToDate: Pernio (Chilbllains), Kroshinsky d, et al Sept. 2025

Panniculitis

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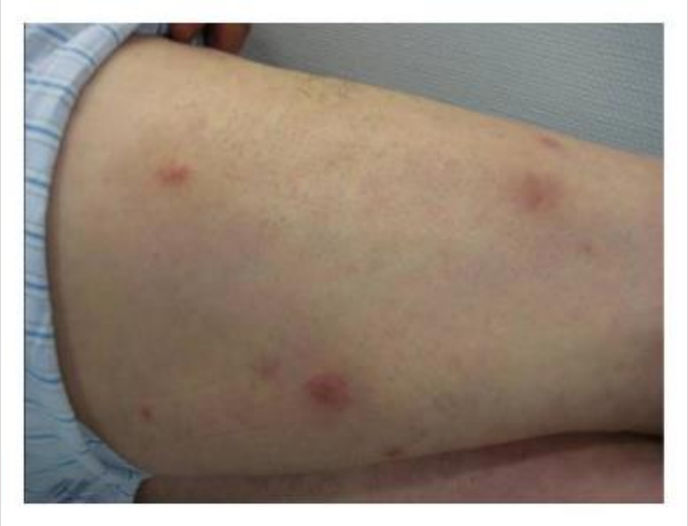
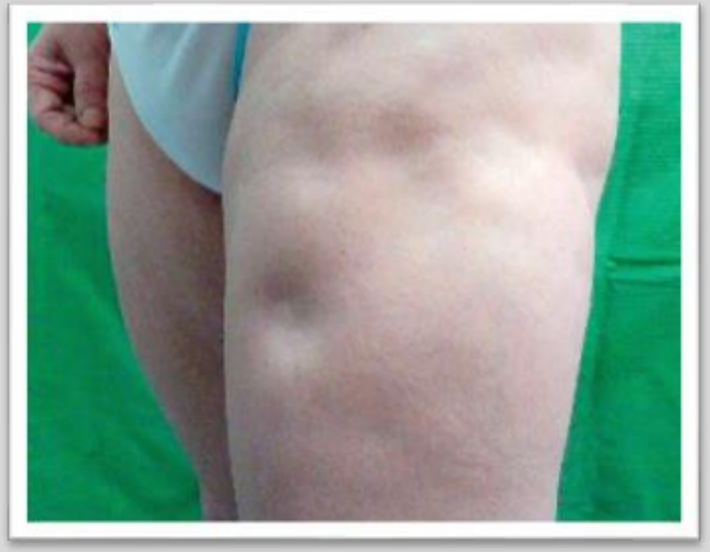
- Long-term, exposure to above-freezing temperatures resulting in necrosis of subcutaneous fat
- Commonly affects equestrian riders
- Long term cosmetic effects
- Rx: Supportive



https://www.researchgate.net/figure/a-Inflammatory-nodules-on-thighs-b-Panniculitis-with-eosinophils-infiltration_fig3_45189062



https://www.semanticscholar.org/paper/Case-report-on-a-patient-with-lupus-panniculitis-Bednarek-Bartoszak/c344f698b4ca23f44638f880c865f78511dfa4a3

Cold Urticaria

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- Just what it sounds like
- Mediated by **histamines**
- Anaphylaxis is rare
- Associated with atopy
- Rx: antihistamines; leukotriene inhibitors; epinephrine

Image from:
http://flipper.diff.org/app/items/info/5339

Frostbite

- When skin temperature reaches **<0°C (Thus, it cannot occur when the ambient temperature is above freezing)**
- Factors: ambient temp; wind; insulation; exposure duration; host factors
- Upon thawing: **arachidonic acid cascade**→ vasoconstriction → platelet aggregation; leukocyte sludging; erythrostasis → thrombosis → ischemia (**manifesting as late pallor or cyanosis**, which helps distinguish this from non-freezing injuries)→ necrosis and dry gangrene

Mainstays of Frostbite Treatment

- **Rapid** rewarming ONLY after re-freezing risk has been eliminated
- **Immersion in, or application of warm water at 37°C to 39°C** (98.6°F to 102.2°F) until affected area is pliable and erythematous;
- Opioid analgesia
- Tetanus immunization status should be assessed and appropriate vaccination administered if needed, because frostbite is a tetanus-prone wound

Frostbite Treatment

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- **Noncontroversial**
  - **Ruptured clear blisters should be debrided** - removes destructive thromboxane and prostaglandins; allows application of topicals.
  - **Hemorrhagic – leave these alone** → worse outcome if debrided (except if very large and over a joint). One study showed you could aspirate*
  - Intravenous/intra-arterial anticoagulation and **thrombolysis for severe frostbite** (1 C recommendation = strong recommendation with weak evidence)**

- **Controversial**
  - Unruptured clear blister debridement – theoretically, blister contents are sterile
  - Systemic antibiotics
  - Topicals - bacitracin; silver sulfadiazine; aloe vera
  - Methylprednisolone
  - Sympathetic blockade –
    - Surgical: sympathectomy
    - Medical: bupivacaine - a long-acting local anesthetic
  - Hyperbarics, heparin and dextran, – all unproven

*Frostbite: pathogenesis and treatment. AUMurphy JV, Banwell PE, Roberts AH, McGrouther DA SOJ Trauma. 2000;48(1):171.

**UpToDate, Frostbite: Acute Care and Prevention, Zafren K. et al, September 2025

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Surgery should be **delayed** to ascertain demarcation (note thumbs).

A

Source: J.E. Tintinalli, J.S. Stapczynski, O.J. Ma, D.M. Yealy, G.D. Meckler, D.M. Cline: Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th Edition
www.accessmedicine.com

B

Source: J.E. Tintinalli, J.S. Stapczynski, O.J. Ma, D.M. Yealy, G.D. Meckler, D.M. Cline: Tintinalli's Emergency Medicine: A Comprehensive Study Guide, 8th Edition
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Disposition

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It depends . . .

If discharged , follow-up with burn center or plastic surgery

Hypothermia = Core Temperature < 35°C

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Image from: https://www.nationalgeographic.com.au/musk-ox/

- <u>Primary hypothermia</u>: a low body temperature caused by a cold environment
- <u>Secondary hypothermia:</u> a low body temperature resulting from medical illnesses, some of which lower the temperature set-point

Causes of Secondary Hypothermia (It's not what this lecture is about, but these exacerbate the problem)

- **Increased Heat Loss**
  - Burns
  - Iatrogenic (i.e., blood transfusions and other cold infusions, cooling blankets, inadequate insulation)
  - Recent birth
- **Impaired Thermogenesis**
  - Impaired shivering (i.e., advanced or very young age, malnutrition, physical exhaustion, neuromuscular disease)
- **Multifactorial**
  - Medications and other xenobiotics i.e., alcohol (COMMON), anesthetic agents, opioids, sedatives, vasodilators)
  - Sepsis especially in elderly or cachectic patients (one of the sepsis criteria is temp. <36°C)
  - Metabolic and endocrine disorders (i.e., alcoholic or diabetic ketoacidosis, adrenal failure, hypoglycemia, hypopituitarism, hypothyroid, lactic acidosis, Wernicke's encephalopathy). These are RARE causes of hypothermia.
  - Neurologic
    - lesions affecting hypothalamus
    - spinal cord injury causing autonomic dysreflexia)
  - Shock
  - Trauma with blood loss

Ability to shiver is crucial in raising body temperature

- Shivering can be suppressed by:
  - Medications (e.g., analgesics, sedatives)
  - Exhaustion of energy stores
  - The application of warming devices (e.g., hot packs; forced-air blankets; warm, humidified air)
  - A core temperature drop below a critical level (~31°C).
    - The suppression of shivering by various warming methods is one of the reasons why the rewarming literature is so controversial.

Physiologic Responses to Cold

- Increased muscle tone and shivering
- Peripheral vasoconstriction
- Cold diuresis – peripheral vasoconstriction leads to elevated BP and thus increased renal blood flow. Kidneys respond by diuresing. In addition, contraction of muscles around bladder causes an earlier urge to void
- Progressive CNS changes:
  - impaired judgment, amnesia, dysarthria, ataxia, diminished consciousness, areflexia (careful with early spinal cord injury, which can also cause areflexia), electroencephalographic arrest, fixed and dilated pupils. *All may be reversible.* No one may be declared dead until they are **warm and dead***

*Nobody is dead until warm and dead": Prolonged resuscitation is warranted in arrested hypothermic victims also in remote areas – A retrospective study from northern Norway, J. Hilmo, et al
resuscitation.2014.04.029

Paradoxical Undressing after Catecholamine Depletion

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Image from:
https://journals.sagepub.com/doi/abs/10.1177/002580248602600310?journalCode=msla

Image from:
https://journals.sagepub.com/doi/abs/10.23907/2018.005?journalCode=afpa

Image from:
https://www.strangeoutdoors.com/questions-answers/tag/What+causes+paradoxical+undressing%3F

Cardiac Changes

- Slow atrial fibrillation
  - Will likely resolve with rewarming – **(pacing and cardioversion are generally NOT indicated)**
  - If tachycardic, consider OTHER causes because bradycardia is the normal response to hypothermia.
- Shivering artifact
- Osborn or "J" Wave (see blue arrow)
- Prolonged QT interval
- Myocardial irritability
  - Handle gingerly. ***Begin CPR only in cardiac arrest.***

Image adapted from: https://www.slideshare.net/venkatatreya/osborn-wave-venkat-ppt

Metabolic Responses

- Early increase in metabolic rate
- Later diminution of metabolism as core temperature drops
- Decreased oxygen demand
  - Hypothetical reason for neuroprotective effect of therapeutic hypothermia
- Rhabdomyolysis
- Pseudo rigor mortis
- Coagulopathy and platelet dysfunction
  - BOTH hypo and hypercoagulability are possible. This is similar to Disseminated Intravascular Coagulation (DIC)

Trauma patients are naked so the room must be kept warm, otherwise they become coagulopathic.

Image from: https://hobbyrobotyka.pl/wp-content/uploads/2016/05/mr_00072158.jpg

Labs

- Glucose is the only lab that **should** be done
- Consider:
  - Creatine phosphokinase (CPK)
  - Electrolytes – cold blood is prone to hemolysis so interpretation is challenging
  - Lactate
  - Blood culture
  - Thyroid stimulating hormone
  - Cortisol
  - Ethanol
  - Osmolal gap
  - Blood gas
    - machine rewarms sample, but do not correct for this – (so called α-stat strategy v. pH-stat strategy)
  - Coagulation factors
    - Unreliable because assay machine rewarms the blood sample

Management

- Depends on CORe temperature and clinical features
- COR = heart
- Measure with a *thermistor* (an electrical device whose resistance varies with temperature) which is more reliable than a thermometer. Make sure it can read extreme temperatures.
- Temperature in the lower esophagus best reflects that of the heart. Rectum and bladder are less reflective.

Management – Stage 1: 35-32°C (conscious and shivering)

- Warm environment and dry clothing
- No hot baths - can cause vasodilation and convective cooling
- Warm, sweet drinks (thermogenesis requires immediate calories)
- Active movement (if possible)
- Thermistor core temperature monitoring not needed

Management – Stage 2: ‹32-28°C (**impaired consciousness**, +/- shivering)

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- Active external (warm environment; chemical, electrical, or forced-air heating; warm packs or warm blankets)
- Minimally invasive rewarming techniques
  - Warm (38-42°C) **parenteral** fluids;

  (airway is not likely to be protected with impaired consciousness, nothing by mouth (NPO-nil per os); warm, humidified oxygen; bladder or nasogastric irrigation with warm fluids.
- Thermistor core temperature monitoring
- Minimal and cautious movements to avoid arrhythmias
- Full-body insulation in the horizontal position
- Consider other causes of impaired consciousness – ETOH, head injury, infection etc.

Management – Stage 3: <28°C (unconscious, BP or pulse still present – so **no** CPR)

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- **Stage 2 management plus:**
- **Airway management as required (NOT contraindicated).**
- **Keep NPO**
- Dysrhythmias are often refractory to medications until patient is rewarmed
  - Medications have altered metabolism at low temperatures. Thus, when rewarmed, there is potential for overdose. Consider **deceased drug dosages** (even though this is controversial, **do not increase dose**)*
- **Preference is to treat** in an **E**xtra**c**orporeal **M**embrane **O**xygenation (ECMO) or **C**ardio-**P**ulmonary **B**ypass (CPB) Center, if available, due to the high risk of cardiac arrest.
  - Especially in cases with cardiac instability that are refractory to medical management
  - Especially in comorbid patients who are unlikely to tolerate low cardiac output
  - Only if ECMO is unavailable should pleural or peritoneal warm saline lavage be considered

*Poloyac M., Empey PE, Drug dosing during hypothermia: to adjust, or not to adjust, that is the question, Pediatr Crit Care Med. 2013 Feb; 14(2): 228–229.

Management – Stage 4: May occur at <32° but usually < 28°C (no BP or pulse present, no forward blood flow on ultrasound)

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- **CPR**
- Up to 3 doses of epinephrine and **defibrillation (if in ventricular fibrillation**). Further dosing is guided by clinical response.
- Airway management
- Transport to ECMO/CPB.
- Only if ECMO is unavailable should pleural or peritoneal warm saline lavage be considered
- Active external WITH minimally invasive rewarming during transport is recommended
  - External rewarming by itself or rewarming the extremities without simultaneously rewarming the trunk MAY cause afterdrop
- do not apply heat to head

Exercise particular care with:

- Sedatives and analgesics – may cause vasodilation, but risk of benefit may outweigh risk of harm (yes, that's the way it should be phrased)

- Vasopressors – patients are likely maximally vasoconstricted early. Their use in later stages is controversial.
  - **Ones with beta1 effects (e.g dobutamine) are preferred over ones with alpha (e.g. norephineprine)**
- Central venous access – keep catheter away from irritable right ventricle.

Afterdrop

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- As periphery warms and vasodilates, warmer core blood is shunted to the periphery
- Studies show it’s minimal*

*Giesbrecht GG, Goheen MS, Johnston CE, Kenny GP, Bristow GK, Hayward JS: Inhibition of shivering increases core temperature afterdrop and attenuates rewarming in hypothermic humans. *J Appl Physiol (1985)* 83: 1630, 1997.

*Giesbrecht GG, Bristow GK: The convective afterdrop component during hypothermic exercise decreases with delayed exercise onset. *Aviat Space Environ Med* 69: 17, 1998.

These are very poor prognostic signs which support the decision to terminate resuscitative efforts:

- Serum **K+ > 12** mmol/L
- **Rewarmed >32°C** and still with no forward blood flow
- Ultrasound shows **no forward flow of blood**? **Vasopressors will not likely have any effect.**
- Obvious mortal injury e.g. decapitation, decomposition
- Witnessed prior normothermic arrest
- Drowning is a bad prognosis. Case reports of survival usually involve very young children with intact diving reflexes.
- Frozen solid
- If none of the above, then RECONSIDER an in place DNR order, because in early stages of hypothermia the prognosis is better

Image from: https://www.ranker.com/list/facts-about-making-the-shining/mike-mcgranaghan

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Sources:

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https://comresearchdata.nyit.edu/redcap/surveys/?s=HRCY448FWYXREL4R